BIOT 6900 Module 3: Target Triage with Protein Language Models

Nicholas Adrogue

Disease: Alzheimer's (AD)

Intro

I used the ESM-C 300M model and a logistic regression prior trained on the panel, and used this for the top 15 module 2 targets. None of the genes were unmapped. None of the panel proteins were removed because none of my targets were in the panel. For the variants, UniProt precursor numbering (APOE R176C = ε2, C130R = ε4; mature R158C and C112R) were used.

1. Re-ranking

The prior from this week mostly agreed with the ranking from week 2. The top stayed mostly consistent with a few changes of about 3 rankings, and one larger ranking change: APOE. APOE moved from rank 1 to rank 6, with a p_sm of 0.002 and a p_ab of 0.016. This means that the model labels APOE as not very similar to small molecule or antibody-drugged proteins, causing the ranking to lower.

The weight I used was 0.65. This was done mainly because the prior was found to be only moderately accurate, with a family-split AUROC of 0.73. Given this, the disease-specific rankings were given a higher ranking.

2. Modality: where p_ab and p_sm differ

One of the proteins with the biggest difference in p_sm and p_ab was HCN1, with p_sm = 0.879 and p_ab = 0.005. HCN1, one of the proteins defined as a hyperpolarization-activated, cyclic nucleotide-gated cation channel [1], is a channel protein on the plasma membrane that allows for sodium and potassium ions to pass through. It is involved in neurological signalling[2].

Using cryogenic-electron microscopy, Lee & MacKinnon resolved the structure of HCN1 with a resolution of 3.5 Å [3]. Che et al. then found the pocket in which ivabradine is bound, which is formed between S1, S4 and the HCN domain. This discovery was supported by evidence in mutagenesis, electrophysiology and molecular dynamics, and demonstrated that this site is

ligandable. Then, using structure-based virtual screening, they found other HCN1 inhibitors that can interact with the same binding site, making this a plausibly druggable target [4]. Given this is a small molecule pocket, the p_sm (0.88) and low p_ab (0.005) are consistent with the biology.

3. Validation: family-split metrics

The family split is a more honest model because of the relationship within protein families. With the random split, this similarity between familial proteins is hidden, and therefore the train and test set are not completely independent of each other. This is because there could be proteins from the same family in both, which are not independent.

The probability values of both AUROC and average precision decrease when using a family split, which is consistent with the idea that prediction accuracy would be inflated with proteins of the same family randomly split between training and test sets.

4. Anchor variants

The score here is a plausibility measure, meaning more positive scores indicate more likely variants to be present evolutionarily and negative scores indicate the opposite. These scores can't see anything regarding the effect the variant actually has, what type of variant it is, or any other context besides likelihood of appearance.

The results support this claim as well. The C130R variant has a high positive score of 7.21, however has been shown to be an important risk factor for AD. Conversely, R176C has a low score of -8.56, but has been shown to be a protector, not a risk factor[5].

5. Limitations and next step

The biggest gap in information from only the sequence is the context of the final protein: its shape and orientation. For this, the 3D view is required.

The first target I will send to investigate further is HCN1. It was ranked number 1 in the listing and has a promising target in the S1/S4/HCN pocket.

The next target I will continue on with is APOE. Though the prior lowered the ranking, I'd like to investigate further to see if the structure of the protein can make up for this going forward.

Sources

[1] Glasscock, E. (2014). Genomic biomarkers of SUDEP in brain and heart. *Epilepsy & Behavior, 38*, 172–179. https://doi.org/10.1016/j.yebeh.2013.09.019

[2] UniProt Consortium. (2026, September 2). *HCN1_HUMAN (O60741): Potassium/sodium hyperpolarization-activated cyclic nucleotide-gated channel 1* (Entry version 193) [Database entry]. UniProtKB. Retrieved October 15, 2026, from https://www.uniprot.org/uniprotkb/O60741/entry

[3] Lee CH, MacKinnon R. Structures of the Human HCN1 Hyperpolarization-Activated Channel. Cell. 2017 Jan 12;168(1-2):111-120.e11. doi: 10.1016/j.cell.2016.12.023. Epub 2017 Jan 12. PMID: 28086084; PMCID: PMC5496774.

[4] Che T, Zhang W, Cheng X, Lv S, Zhang M, Zhang Y, Yang T, Nan W, Wan S, Zeng B, Li J, Xiong B, Zhang J. Structural mechanism of human HCN1 hyperpolarization-activated channel inhibition by ivabradine. J Biol Chem. 2024 Nov;300(11):107798. doi: 10.1016/j.jbc.2024.107798. Epub 2024 Sep 20. PMID: 39307309; PMCID: PMC11530593.

[5] Farrer LA, Cupples LA, Haines JL, Hyman B, Kukull WA, Mayeux R, Myers RH, Pericak-Vance MA, Risch N, van Duijn CM. Effects of age, sex, and ethnicity on the association between apolipoprotein E genotype and Alzheimer disease. A meta-analysis. APOE and Alzheimer Disease Meta Analysis Consortium. JAMA. 1997 Oct 22-29;278(16):1349-56. PMID: 9343467.